Set Theory

Definition

<u>Definition(Sets):</u>

A set is any well-defined list, collection, or class of objects. The objects in a set are called the elements or members, of the set.

A set is said to contain its elements.

The statement "p is an element of A" is written $p \in A$.

The statement "p is not an element of A" is written $p \notin A$.

There are several ways to describe a particular set.

Definition (Cont')

List its members. (If possible)

Example:

$A = \{a, e, i, o, u\}$,

$B = \{a$, 2, New Jersey$\}$

State those properties which characterize the elements in the set.

Example:

$A = \{x \mid x$ is an integer, $x > 0\}$

$B = \{x \mid x$ is an odd positive integer less than 12$\}$

Note:

We often use this type of notation to describe sets when it is impossible to list all the elements of the set.

For example, R = $\{x \mid x$ is a real number$\}$.

Definition (Cont')

Using Venn diagrams.

- In Venn diagrams, the universal set *U*, which contains all the objects under consideration, is represented by a rectangle.
- Inside this rectangle, circles or other geometrical figures are used to represent sets.
- Sometimes points are used to represent the particular elements of the set.

<u>Example:</u>

- Draw a Venn diagram that represents *Z* , the set of integers, in the Real numbers set *R* .

<u>Note:</u>

Venn diagrams are often used to indicate the relationships between sets.

Finite and infinite sets

Definition:

Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a finite set and n is the cardinality of S.

The cardinality of S is denoted by |S|.

Definition:

A set is said to be infinite if it is not finite.

Example:

(i)	The set of the days of the week.	Finite
(ii)	N = {2, 4, 6, 8,…}.	Infinite
(iii)	P = {x| x is a river on the earth}	Infinite
(iv)	$\{x \in R \mid -2 < x < 5\}$	Infinite
(v)	$\{x \in Z \mid -2 < x < 5\}$	Finite
(vi)	$\{x \in R+ \mid -2 < x < 5\}$	Infinite

Equality of Sets

Definition:

Set A is equal set B if they both have the same members. We denote the equality of sets A and B by A = B.

Example:

(i) $\{1, 2, 3, 4\} = \{3, 1, 4, 2\}$? Equal

(ii) $\{5, 6, 5, 7\} = \{7, 5, 7, 6\}$? Not Equal

(iii) $\{1, 3, 3, 3, 5, 5, 5, 5\} = \{1, 3, 5\}$? Not Equal

(iv) $\{x \mid x2 - 3x = -2\} = \{1, 2\}$? Equal

Equality of Sets

(v) Let the sets A, B, C and D be defined as follows:

A = {n ∈ Z | n = 2p for some integer p} B = the set of all even integers

C = {m ∈ Z | m = 2q – 2 for some integer q} D = {k ∈ Z | k = 3r + 1 for some integer r}

A = B? A = C? A = D?

A = {K, − 4, − 2, 0, 2, 4, K}

B = {K, − 4, − 2, 0, 2, 4, K}

C = {K, − 4, − 2, 0, 2, 4, K}

D = {K, − 5, − 2, 1, 4, 7, 10, K}

A = B , A = C ,A ≠ D

Empty Set

Definition:

A set which contains no elements is called the empty set (or null set), and is denoted by ∅.

Example:

$\{x \mid x^2 = 4, x \text{ is odd}\}$ is the empty set.

$\{x \in \mathrm{R} \mid x^2 = -1\}$ is the empty set.

Subsets

Definition:

If A and B are sets, A is called a subset of B, written $A \subseteq B$, if and only if every element of A is also an element of B.

More specifically, A is a subset of B if $x \in A$ implies $x \in B$.

The phrases “A is contained in B” and “B contains A” are alternative ways of saying that A is a subset of B.

Subsets (Cont')

Example:

(a) Suppose B76, XR3, D54, ES2 and XL5 are the model numbers of certain pieces of equipment.

Let A = {B76, XR3, D54, ES2}, B = {B76, D54} and C = {ES2, XL5}.

$B \subseteq A$?	$C \subseteq A$?	$B \subseteq B$?
$B \subseteq A$	$C \nsubseteq A$	$B \subseteq B$

(b) {x | x is a positive power of 2} ⊆ {x | x is even}?

{x | x is a positive power of 2} = {21 , 22 , 23 , 24 , ...}

{x | x is even} = {..., − 4, − 2, 0, 2, 4, 6, 8,...}

{x | x is a positive power of 2} ⊆ {x | x is even}

Subsets (Cont')

Theorem:

For any set A,

(i) $\emptyset \subseteq A$ and (ii) $A \subseteq A$.

**Note:

If A is not a subset of B, that is, written $A \nsubseteq B$, then there is at least one element in A that is not a member of B.

<u>Transitive property of subsets</u>

For all sets A, B and C, if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Proper Subset

Definition:

Let A and B be two sets. A is a proper subset of B ($B \subset A$) if and only if every element of A is in B but there is at least one element of B that is not in A.

Example:

(i) Let A = {x | x is a multiple of 4}, B = {x | x is a multiple of 8}. Show that $B \subset A$.

Let $x \in B \Rightarrow x = 8k$ for some integer k.

$= 2(4)k$

$= 4m$ where $m = 2k$

$\therefore$ x is a multiple of 4

$\therefore x \in A$

$12 \in A$ but $12 \notin B$

$\therefore B \subset A$

Proper Subset (Cont')

(ii) Let $A = \{x \mid x \in R \text{ and } x2 - 4x + 3 = 0\}$

$B = \{x \mid x \in N \text{ and } 1 \leq x \leq 4\}$

Show that $A \subset B$.

$A = \{x \mid x \in R \text{ and } x2 - 4x + 3 = 0\} \Rightarrow A = \{1, 3\}$

$B = \{x \mid x \in N \text{ and } 1 \leq x \leq 4\} \Rightarrow B = \{1, 2, 3, 4\}$

$\therefore A \subset B$.

Proper Subset (Cont')

<u>Proper Subsets</u> - For Sets *A* and *B*, Set *A* is a **Proper Subset** of Set *B* if every element in Set *A* is also in Set *B*, ***but*** *<u>Set A does **not** equal Set B.</u>* $(\boldsymbol{A} \neq \boldsymbol{B})$ It is written as $\boldsymbol{A} \subset \boldsymbol{B}$.

Example: $Set\ A = \{2,4,6\}$ $Set\ B = \{0,2,4,6,8\}$

$\{2,4,6\} \subseteq \{0,2,4,6,8\}$ **and** $\{2,4,6\} \subset \{0,2,4,6,8\}$

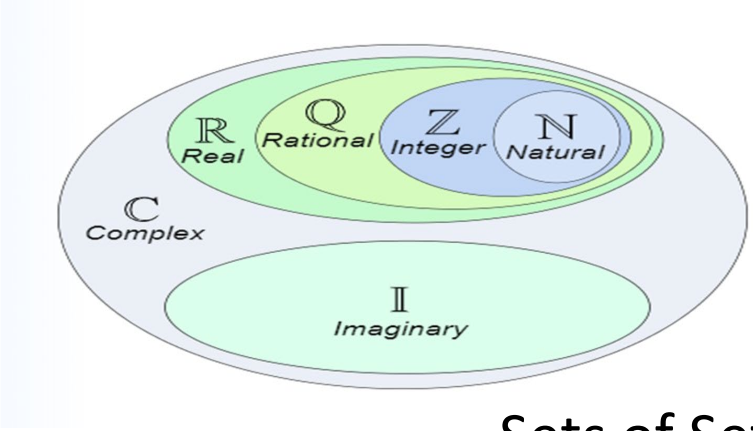
Set *A* is a **<u>Subset</u>** of Set *B* because every element in *A* is also in *B*. $\boldsymbol{A} \subseteq \boldsymbol{B}$

Set *A* is a **<u>Proper Subset</u>** of Set *B* because every element in *A* is also in B, but A $\neq$ *B*. $\boldsymbol{A} \subset \boldsymbol{B}$

Note: *The Empty Set is a Subset of every Set.*
The Empty Set is also a Proper Subset of every Set <u>except</u> the Empty Set.

Relations among sets of numbers

Let Z, Q and R denote the sets of integers, rational numbers and real numbers respectively.



- Z = the set of integers
- Q = the set of rational numbers
- R = the set of real numbers
- Z ⊂ Q since every integer is rational. (½ ∈ Q)
- Q ⊂ R since every rational number is real (√2 ∈ R but √2 ∉ Q)

<u>Sets of Sets (Or Family of sets)</u>

Sometimes it will happen that the objects of a set are sets themselves. For example, {{2,3}, {2}, {5, 6}} is a family of sets.

Power Set

Definition:

Given a set S, the power set of S, P(S), is the set of all subsets of the set S.

Example:

(i) What is the power set of the set {0, 1, 2}?

$$P(S) = \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}, \phi\}$$

(ii) What is the power set of the set {a, b}?

$$P(S) = \{a\}, \{b\}, \{a, b\}, \phi\}$$

(iii) What is the power set of the empty set? (i.e. $A = \phi$)

$$P(S) = \{\phi\}$$

(iv) What is the power set of the set $\{\phi\}$? (i.e. $A = \{\phi\}$)

$$P(S) = \{\phi, \{\phi\}$$

What is a Power Set?

In set theory, the power set (or power set) of a Set A is defined as the set of all subsets of the Set A including the Set itself and the null or empty set. It is denoted by P(A). Basically, this set is the combination of all subsets including null set, of a given set.

How is Power set Calculated?

If the given set has n elements, then its Power Set will contain 2^n elements. It also represents the cardinality of the power set.

Example of Power Set

Let us say Set A = { a, b, c }

Number of elements: 3

Therefore, the subsets of the set are:

- { } which is the null or the empty set
- { a }
- { b }
- { c }
- { a, b }
- { b, c }
- { c, a }
- { a, b, c }

The power set P(A) = { { } , { a }, { b }, { c }, { a, b }, { b, c }, { c, a }, { a, b, c } }

Power Set (Con't)

Theorem:

For all sets A and B, if $A \subseteq B$, then $P(A) \subseteq P(B)$.

$x \in P(A) \Rightarrow x \in A$ and $A \subseteq B \Rightarrow x \in B$

$\therefore x \in P(B)$

$\therefore P(A) \subseteq P(B)$

Theorem:

For all integers $n \geq 0$, if a set X has n elements, then P(X) has 2^n elements.

Power Set (Con't)

Proof. ($\Rightarrow$) Suppose $A \subseteq B$. We need to show that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Take any $X \in \mathcal{P}(A)$. Then $X \subseteq A$. Since $A \subseteq B$, we have by Proposition 3.3.1 that $X \subseteq B$. Therefore $X \in \mathcal{P}(B)$. Therefore $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

($\Leftarrow$) Suppose $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. We need to show that $A \subseteq B$.

Since $A \subseteq A$, $A \in \mathcal{P}(A)$. Since $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, $A \in \mathcal{P}(B)$. By definition of $\mathcal{P}(B)$, $A \subseteq B$. $\square$

Distinction between ∈ and ⊆

Distinction between ∈ and ⊆:

Example:

Which of the following are true statements?

(i)	$2 \in \{1, 2, 3\}$ TRUE	(iv)	$\{2\} \in \{1, 2, 3\}$	FALSE
(ii)	$2 \subseteq \{1, 2, 3\}$ FALSE	(v)	$\{2\} \subseteq \{1, 2, 3\}$	TRUE
(iii)	$\{2\} \subseteq \{\{1\}, \{2\}$ TRUE	(vi)	$\{2\} \in \{\{1\}, \{2\}\}$	TRUE

Cartesian Products

Definition:

Let A and B be sets. The Cartesian product of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Example:

What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$? Is $A \times B = B \times A$?

$$A \times B = \{ (1, a), (1, b), (1, c), (2, a), (2, b), (2, c) \}$$
$$B \times A = \{ (a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2) \}$$

$$\therefore A \times B \neq B \times A$$

Basic Set Operations(Union)

Definition (Union)

The union of sets A and B is the set of all elements which belong to A or to B or to both, it is denoted by $A \cup B$.

Example:

(i) Let $S = \{a, b, c, d\}$ and $T = \{f, b, d, g\}$.

Then $S \cup T = \{a, b, c, d, f, g\}$

(ii) Let P = the set of positive real numbers

Q = the set of negative real numbers.

Then $P \cup Q = \{x \mid x \in R, x \neq 0\}$

Basic Set Operations(Con't)

(iii) R = the set of all real no = universal set.

$A = \{x \in R \mid -1 < x \leq 0\}$

$B = \{x \in R \mid 0 \leq x < 1\}$

Then $A \cup B = \{x\ x \in R, -1 < x < 1\}$

Note:

(1) The union of A and B may be defined concisely by

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

(2) $A \cup B = B \cup A$.

(3) $A \subseteq (A \cup B)$ and $B \subseteq (A \cup B)$.

Basic Set Operations (Intersection)

<u>Definition(Intersection)</u>

The intersection of sets A and B is the set of elements which are common to A and B, it is denoted by $A \cap B$.

Example:

(i) Let $S = \{a, b, c, d\}$ and $T = \{f, b, d, g\}$. Then $S \cap T = \{b, d\}$

(ii) Let $V = \{2, 4, 6, 8, \ldots\}$ and $W = \{3, 6, 9, 12, \ldots\}$.

Then $V \cap W = \{6, 12, 18, K\}$

Basic Set Operations (Intersection)(Cont')

(iii) R = the set of all real no = universal set.

$A = \{x \in R \mid -1 < x \leq 0\}$

$B = \{x \in R \mid 0 \leq x < 1\}$

Then $A \cap B = \{0\}$

Note:

(1) The intersection of A and B may be defined concisely by

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

(2) $A \cap B = B \cap A$.

(3) $(A \cap B) \subseteq A$ and $(A \cap B) \subseteq B$.

(4) If sets A and B have no elements in common (A and B are disjoint), then $A \cap B = \emptyset$.

Basic Set Operations (Difference)

Definition(Difference)

The difference of sets A and B is the set of elements which belong to A but do not belong to B, it is denoted by A – B.

Example:

(i) Let S = {a, b, c, d} and T = {f, b, d, g},

then S – T = {a, c}

(ii) Let R = the set of real numbers and Q = the set of rational numbers,

then R – Q = the set of irrational numbers

Basic Set Operations (Difference)(Cont')

Note:

(i) The difference of A and B may be defined concisely by

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}.$$

(ii) $(A - B) \subseteq A$.

Let $x \in (A - B) \cap B \Rightarrow x \in A - B$ and $x \in B$

$\Rightarrow x \in A$ and $x \in B$ and $x \notin B$

$\therefore (A - B) \cap B = \emptyset$

(iii) The set $A - B$, $A \cap B$ and $B - A$ are mutually disjoint.

$(A - B) \cap (A \cap B) = \emptyset$, $(A - B) \cap (B - A) = \emptyset$ and $(A \cap B) \cap (B - A) = \emptyset$

$x \in (A - B) \cap (A \cap B) \Rightarrow x \in A - B$ and $x \in A \cap B$

$\Rightarrow x \in A$ and $x \notin B$; and $x \in A$ and $x \in B$

(iii) Cont'

No element will satisfy both $x \notin B$ and $x \in B$

$\therefore (A - B) \cap (A \cap B) = \emptyset$

$x \in (A - B) \cap (B - A) \Rightarrow x \in A - B$ and $x \in B - A$

$\Rightarrow x \in A$ and $x \notin B$; and $x \in B$ and $x \notin A$

No element will satisfy both $x \notin B$ and $x \in B$, $x \in A$ and $x \notin A$

$\therefore (A - B) \cap (B - A) = \emptyset$

Basic Set Operations (Complement)

Definition(Complement)

The complement of a set A is the set of elements which do not belong to A, it is denoted by A^c.

Example:

(i) Let the universal set U be the English alphabet and let T = {a, b, c}.

Then $T^c = \{d, e, f, g, K, z\}$

(ii) Let $E = \{-4, -2, 0, 2, 4, 6, \ldots\}$. Then $E^c = \{-3, -1, 1, 3, 5, 7, \ldots\}$.

Operations on Comparable sets

Definition:

Two sets A and B are called comparable if $A \subset B$ or $B \subset A$.

The operations of union, intersection, difference and complement have simple properties when the sets under investigation are comparable.

(i) $A \subset B$ implies $A \cap B = A$.

(ii) $A \subset B$ implies $A \cup B = B$.

(iii) $A \subset B$ implies $Bc \subset Ac$.

(iv) $A \subset B$ implies $A \cup (B – A) = B$.

1.1.8 Comparability:

The set A and B are said to be comparable if $A \subset B$ or $B \subset A$.

Two sets A and B are said to be not comparable if $A \not\subset B$ and $B \not\subset A$.

<u>Example:</u> Let A = {a, b}, B = {a, b, c} and C = {a, d}. Then A is comparable to B, since $A \subset B$ and B and C are not comparable to each other since $C \not\subset B$ and $B \not\subset C$.

Set Identities

1) Associative laws

For all sets A, B and C,

$$(A \cap B) \cap C = A \cap (B \cap C) \text{ and } (A \cup B) \cup C = A \cup (B \cup C).$$

2) Distributive laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \text{ and}$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

3) Intersection with U (U acts as an identity for intersection)

$$A \cap U = A.$$

4) Idempotent laws

$$A \cap A = A \text{ and } A \cup A = A.$$

Set Identities(Cont')

5) De Morgan's laws

$(A \cup B)^c = Ac \cap B^c$ and $(A \cap B)^c = Ac \cup B^c$.

6) Union with U (U acts as a universal bound for union)

$A \cup U = U$.

7) Absorption laws

$A \cup (A \cap B) = A$ and $A \cap (A \cup B) = A$.

8) Intersection with Ø (Ø acts as a universal bound for intersection)

$A \cap \emptyset = \emptyset$.

Suppose $A \cap \emptyset \neq \emptyset$.

$\therefore \exists$ x such that $x \in A \cap \emptyset$

Then $x \in A$ and $x \in \emptyset$

But $x \in \emptyset$ is not possible since Ø has no elements So $A \cap \emptyset = \emptyset$.

Basic method for proving that sets are equal To prove $A = B$

(i) prove that $A \subseteq B$ and
(ii) prove that $B \subseteq A$

Example:

(i) Prove that $\{x \mid x \in N \text{ and } x2 < 15\} = \{x \mid x \in N \text{ and } 2x < 7\}$.

Let $A = \{x \mid x \in N \text{ and } x2 < 15\} = \{1, 2, 3\}$

$B = \{x \mid x \in N \text{ and } 2x < 7\} = \{1, 2, 3\}$

So $A = B$.

(ii) Prove that for all sets A, B and C,

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Let A, B and C be sets Suppose $x \in A \cup (B \cap C)$

$\therefore x \in A$ or $x \in (B \cap C)$

Basic method for proving that sets are equal To prove $A = B$

Case 1

$x \in A \Rightarrow x \in A \cup B$

$\Rightarrow x \in A \cup C$

$\Rightarrow x \in (A \cup B) \cap (A \cup C)$

Case 2

$x \in (B \cap C) \Rightarrow x \in B$ and $x \in C$

$\Rightarrow x \in A \cup B$ and $x \in A \cup C$

$\Rightarrow x \in (A \cup B) \cap (A \cup C)$

$\Rightarrow A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$

Suppose $x \in (A \cup B) \cap (A \cup C)$

$\therefore x \in (A \cup B)$ and $x \in (A \cup C)$

Case 1

$x \in A \Rightarrow x \in A \cup (B \cap C)$

Case 2

$x \notin A$

$x \in (A \cup B) \Rightarrow x \in B$

$x \in (A \cup C) \Rightarrow x \in C$

Hence $x \in B \cap C$

So $x \in A \cup (B \cap C)$

So $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$

$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$ and $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$

$\Rightarrow A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Basic method for proving that sets are equal To prove $A = B$

(iii) Prove that for all sets A and B,

$$(A \cup B)^c = Ac \cap B^c.$$

Let A and B be sets Suppose $x \in (A \cup B)^c$

So $x \notin A \cup B \Rightarrow x \notin A$ and $x \notin B$

$\Rightarrow x \in A^c$ and $x \in B^c$

$\Rightarrow x \in A^c \cap B^c$

$(A \cup B)^c \subseteq A^c \cap B^c$

Suppose $x \in A^c \cap B^c \Rightarrow x \in A^c$ and $x \in B^c$

$\Rightarrow x \notin A$ and $x \notin B$

$\Rightarrow x \notin A \cup B$

$\Rightarrow x \in (A \cup B)^c$

$A^c \cap B^c \subseteq (A \cup B)^c$

So $(A \cup B)^c \subseteq A^c \cap B^c$ and $A^c \cap B^c \subseteq (A \cup B)^c \Rightarrow A^c \cap B^c = (A \cup B)^c$

Basic method for proving that sets are equal To prove A = B

(iv) Prove that for all sets A, B and C,

$(A \cup B) - C = (A - C) \cup (B - C)$.

$(A \cup B) - C \quad = (A \cup B) \cap Cc$

$= Cc \cap (A \cup B)$

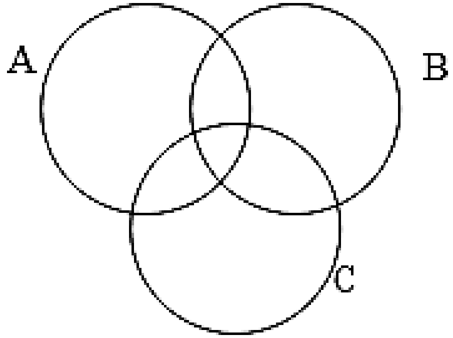
$= (Cc \cap A) \cup (Cc \cap B)$

$= (A \cap Cc) \cup (B \cap Cc)$

$= (A - C) \cup (B - C)$

(v) $(A - B) \cup (B - C) = A - C$, true or false?

(i) using Venn diagram



$\therefore$ *False*

(ii) using counterexample:

A = {a, b}, B = {b, c} and C = {a, d} A – B = {a}

B – C = {b, c}

A – C = {b}

So (A – B) ∪ (B – C) = {a, b, c} Hence (A – B) ∪ (B – C) ≠ A – C

Disjoint

Definition:

Two sets are called disjoint if and only if they have no elements in common.

Example:

(i) Are $\{1, 3, 5\}$ and $\{2, 4, 6\}$ disjoint? YES

(ii) Given any two sets A and B, are $A - B$ and B disjoint?

Suppose $(A - B) \cap B \neq \emptyset$

Then $\exists$ x such that $x \in (A - B) \cap B$

$\Rightarrow x \in A - B$ and $x \in B$

$\Rightarrow x \in A$ and $x \notin B$; and $x \in B$

$\therefore x \notin B$; and $x \in B$, a contradiction

$\therefore$Therefore, $A - B$ and B are disjoint.

Mutually Disjoint

Definition:

The sets $A_1, A_2, \dots, A_n$ are mutually disjoint (or pairwise disjoint) if and only if no two sets A_i and A_j with distinct subscripts have any elements in common.

Example:

(i) Let A = {3, 5}, B = {1, 4, 6} and C = {2}. Are A, B and C mutually disjoint?

Since

$$A \cap B = \phi,\ A \cap C = \phi,\ B \cap C = \phi,\ A \cap B \cap C = \phi$$

$\therefore$ A, B, and C are mutually disjoint.

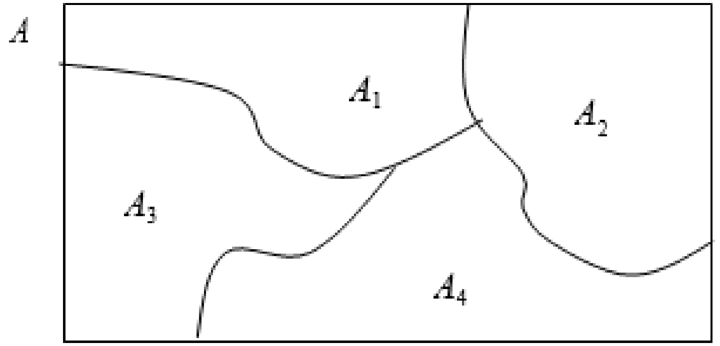
(ii) Let X = {2, 4, 6}, Y = {3, 7} and Z = {4, 5}. Are X, Y and Z mutually disjoint?

Since $X \cap Z \neq \phi$

$\therefore$ X , Y , and Z are NOT mutually disjoint.

Mutually Disjoint (Cont')

Suppose A, A_1, A_2, A_3 and A_4 are the sets of points represented by the regions as shown in the below figure.



- The A_1, A_2, A_3 and A_4 are subsets of A, and $A = A_1 \cup A_2 \cup A_3 \cup A_4$.
- Suppose boundaries are assigned to the regions representing A_1, A_2, A_3 and A_4 in such a way that these sets are mutually disjoint.
- Then A is called a union of mutually disjoint subsets, and the collection of sets $\{A_1, A_2, A_3, A_4\}$ is said to be a partition of A.

Partitions of Sets

Definition:

A collection of nonempty sets $\{A_1, A_2, \ldots, A_n\}$ is a partition of a set A if and only if

(i) $A = A_1 \cup A_2 \cup \ldots \cup A_n$

(ii) $A_1, A_2, \ldots, A_n$ are mutually disjoint.

Example:

Let $A = \{1, 2, 3, 4, 5, 6\}$, $X = \{1, 2\}$, $Y = \{3, 4\}$ and $Z = \{5, 6\}$.

Is $\{X, Y, Z\}$ a partition of A?

$A = X \cup Y \cup Z$

$X \cap Y = \emptyset$ $X \cap Z = \emptyset$ $Y \cap Z = \emptyset$

So $\{X, Y, Z\}$ is a partition of A.

Partitions of Sets (Cont')

Example:

Let Z = the set of all integers

$F = \{n \in Z \mid n = 3k \text{ for some integer } k\}$

$G = \{n \in Z \mid n = 3k + 1 \text{ for some integer } k\}$ $H = \{n \in Z \mid n = 3k + 2 \text{ for some integer } k\}$ Is $\{F, G, H\}$ a partition of Z?

Yes. By the quotient-remainder theorem every integer n can be represented in exactly one of the three forms

$n = 3k$ or $n = 3k + 1$ or $n = 3k + 2$

for some integer k. This implies that no integer can be in any two of the sets F, G or H. So F, G and H are mutually disjoint. It also implies that every integer is in one of the sets F, G or H. So $Z = F \cup G \cup H$.

Russell's Paradox

<u>Russell's Paradox</u>

Most sets are not elements of themselves. However, some sets are elements of themselves.

Let S be the set of all sets that are not elements of themselves. Is S an element of itself?

The answer is neither yes nor no.

For if $S \in S$, then S satisfies the defining property for S, and hence $S \notin S$. But if $S \notin S$, then S is a set such that $S \notin S$ and so S satisfies the defining property for S, which implies that $S \in S$. Thus neither is $S \in S$ nor is $S \notin S$, which is a contradiction.

The Barber Puzzle

In a certain town, there is a male barber who shaves all those men, and only those men, who do not shave themselves. Does the barber shave himself?

Neither yes nor no.

If the barber shaves himself, he is a member of the class of men who shave themselves. But no member of this class is shaved by the barber, and so the barber does not shave himself.

On the other hand, if the barber does not shave himself, he belongs to the class of men who do not shave themselves. But the barber shaves every man in this class, so the barber does shave himself.

The Barber Puzzle(Con't)

- A condition of the puzzle is that the barber is a man who shaves all those men who do not shave themselves. If he does not shave then he does not shave himself, in which case he is shaved by the barber and the contradiction is as present as ever. Similarly, other attempts as resolution of the paradox by considering details of the barber's situation are doomed to failure.

- So let's accept the fact that the paradox has no easy resolution and see where that thought leads. Since the barber neither shaves himself nor doesn't shave himself, the sentence "The barber shaves himself' is neither true nor false. But the sentence arose in a natural way from a description of a situation. If the situation actually existed, then the sentence would have to be true or false. So we are forced to conclude that the situation described in the puzzle simply cannot exist in the world as we know it.

The Barber Puzzle(Con't)

- In a similar way the conclusion to be drawn from Russell's paradox itself is that the object *S* is not a set. Because if it actually were a set, in the sense of satisfying the general properties of sets we have been assuming, then it either would be an element of itself or not.

- In the years following Russell's discovery, several ways were found to define the basic concepts of set theory so as to avoid his contradiction. The way used in this text requires that, except for the power set whose existence is guaranteed by an axiom, whenever a set is defined using a predicate as a defining property, the stipulation must also be made that the set is a subset of a known set. This method does not allow us to talk about "the set of all sets that are not elements of themselves." We can only speak of "the set of all sets that are subsets of some known set and that are not elements of themselves." When this restriction is made, Russell's paradox ceases to be contradictory. Here is what happens:

The Barber Puzzle(Con't)

Let U be a set of sets and let $S = \{A \in U \mid A \notin A\}$. Is $S \in S$?

<u>Solution:</u>

The answer is simply no. For if $S \in S$, then S satisfies the defining property for S, and hence $S \notin S$.

If you have some experience programming computers, you know how badly an infinite loop can tie up a computer system. It would be useful to be able to preprocess a program and its data set by running it through a checking program that determines whether execution of the given program with the given data set would result in an infinite loop. Can an algorithm for such a program be written? In other words, can an algorithm be written that will accept any algorithm *X* and any data set *D* as input and will then print "halts" or "loops forever" to indicate whether *X* terminates in a finite number of steps or loops forever when run with data set *D*? In the 1030s Turing proved that the answer to this question is no.

The Halting Problem

Theorem:

There is no computer algorithm that will accept any algorithm *X* and data set *D* as input and then will output “halts” or “lops forever” to indicate whether *X* terminates in a finite number of steps when *X* is run with data set *D*.

Computer Representation of Set

There are various ways to represent sets using a computer.

Store the elements of the set in an unordered fashion. If this is done, the operations of computing the union, intersection, or difference of two sets would be time-consuming, since each of these operations would require a large amount of searching for elements.

Store the elements using an arbitrary ordering of the elements of the universal set. This method of representing sets makes computing combinations of sets easy.

Assuming that the universal set U is finite (and of reasonable size so that the number of elements of U is not larger than the memory size of the computer being used). First, specify an arbitrary ordering of the elements of U, for instance a_1, a_2, …, a_n. Represent a subset A of U with the bit string of length n, where the ith bit in this string is 1 if a_i belongs to A and is 0 if a_i does not belong to A.

Computer Representation of Set (Con't)

Example:

Let U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, and the ordering of elements of U has the elements in increasing order; i.e., $a_i = i$. What bit strings represent the subset of all odd integers in U, the subset of all even integers in U, and the subset of integers not exceeding 5 in U?

The bit string that represents the set of odd integers in U, namely, {1, 3, 5, 7, 9}, has a one bit in the first, third, fifth, seventh, and ninth positions, and a zero elsewhere. It is

10 1010 1010

(We have split this bit string of length ten into blocks of length four for easy reading since long bit strings are difficult to read.) Similarly, we represent the subset of all even integers in U, namely, {2, 4, 6, 8, 10}, by string

01 0101 0101

The set of all integers in U that do not exceed 5, namely, {1, 2, 3, 4, 5}, is represented by the string

11 1110 0000

Computer Representation of Set (Con't)

Using bit strings to represent sets, it is easy to find complements of sets and unions, intersections, and differences of sets. To find the bit string for the complement of a set from the bit string for that set, we simply change each 1 to 0 and each 0 to 1, since $x \in A$ if and only if $x \notin A^c$. Note that this operation corresponds to taking the negation of each bit when we associate a bit with a truth value – with 1 representing true and 0, false.

<u>Note:</u>

To find the bit string for the complement of a set from the bit string for that set, we simply change each 1 to 0 and each 0 to 1.

Computer Representation of Set(Con't)

<u>Example:</u>

What is the bit string for the complement of the set {1, 3, 5, 7, 9} with universal set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}?

01 0101 0101

<u>Note:</u>

The bit in the *i*th position of the bit string of the union is 1 if either of the bits in the *i*th position in the two strings is 1 (or both are 1), and is 0 when both bits are 0.

The bit in the *i*th position of the bit string of the intersection is 1 when the bits in the corresponding position in the two strings are both 1, and is 0 when either of the two bits is 0 (or both are).

Computer Representation of Set (Con't)

Example:
The bit strings for the sets {1, 2, 3, 4, 5} and {1, 3, 5, 7, 9} are
11 1110 0000 and 10 1010 1010
respectively. Use bit strings to find the union and intersection of these sets

The bit string for the union of these sets is
$11\ 1110\ 0000 \vee 10\ 1010\ 1010 = 11\ 1110\ 1010$
which corresponds to the set {1, 2, 3, 4, 5, 7, 9}

The bit string for the intersection of these sets is
$11\ 1110\ 0000 \wedge 10\ 1010\ 1010 = 10\ 1010\ 0000$
which corresponds to the set {1, 3, 5}

Countably Infinite

<u>Definition:</u>

A set is called countably infinite if, and only if, it has the same cardinality as the set of positive integers Z^+. A set is called countable if, and only if, it is finite or countably infinite. A set that is not countable is called uncountable.

<u>Example:</u>

Z = the set of all integers, is countable.

2Z = the set of all even integers, is countable.